

Topology-Dependent Screening in the PPP Model: A Model for C_{60}

Zahra Karimi

Institute of Physics, Faculty of Physics, Astronomy and
Informatics,

Nicolaus Copernicus University in Toruń

Supervisor: Dr. hab. Paweł Tecmer, prof. UMK

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Pariser–Parr–Pople (PPP) Model Hamiltonian

$$\hat{H}_{\text{PPP}} = \sum_{\langle i,j \rangle, \sigma} t_{ij} \left(\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \right) + \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\substack{\langle i,j \rangle \\ i \neq j}} V_{ij} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

Definitions:

- t_{ij} : nearest-neighbor hopping (kinetic energy term)
- V_{ij} : distance-dependent intersite Coulomb repulsion
- U : on-site Coulomb interaction
- $\hat{c}_{i\sigma}^\dagger, \hat{c}_{i\sigma}$: electron creation and annihilation operators at site i with spin σ
- $\hat{n}_{i\sigma} = \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma}$: spin-resolved number operator
- $\hat{n}_i = \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{n}_{i\sigma}$: total electron number operator at site i

Ohno formula

$$V_{ij} = \frac{U}{\kappa_{ij} \sqrt{1 + 0.6117 R_{ij}^2}}$$

Parameters:

- U : on-site electron-electron repulsion
- κ_{ij} : dielectric screening constant
- R_{ij} : distance between i -th and j -th carbon atoms

Two sets of Coulomb parameters used in literature:

- **Screened parameters:**

- $U = 8.0 \text{ eV}$
- $\kappa_{ij} = 2.0 \quad (i \neq j)$

- **Standard parameters:**

- $U = 11.13 \text{ eV}$
- $\kappa_{ij} = 1.0 \quad (i \neq j)$

What is Topology-Dependent Screening?

In π -conjugated systems:

- Carbon atoms are connected in different ways
- This defines the **molecular topology**

Topology-dependent screening:

- Electron–electron interaction depends on local connectivity
- Screening is not the same for all atoms

Physical idea:

- More connected atoms $\Rightarrow$ more delocalization
- More delocalization $\Rightarrow$ stronger screening

Why Topology-Dependent Screening?

Standard PPP model:

- Uses fixed Coulomb parameters for all atoms
- Assumes identical local environments

Problem:

- Real π -systems are not uniform
- Edge, fused, and internal atoms behave differently
- Local topology affects electron delocalization

Consequence:

- Fixed parameters cannot capture realistic screening
- Leads to errors in excitation energies for real molecules.

Chandross & Mazumdar, Phys. Rev. B (1997)

Topology-Dependent Screening Factor

Local descriptor for each atom:

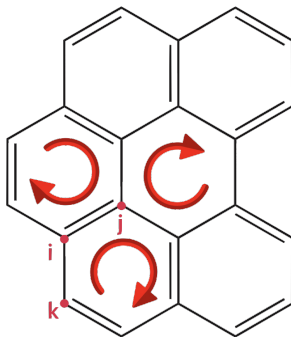
- $n_{\text{loop},i}$, the number of aromatic rings it belongs to
- $N_{\text{CC},i}$, the number of π -connections (equivalent to the adjacency matrix degree)

Local screening weight:

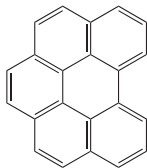
$$s_i = n_{\text{loop},i} \cdot N_{\text{CC},i} \quad (1)$$

Screening between two atoms:

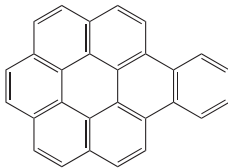
$$k_{ij} = \frac{s_i + s_j}{2} \quad (2)$$



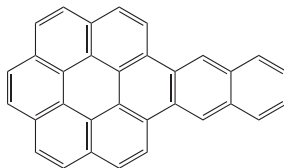
Studied Molecules



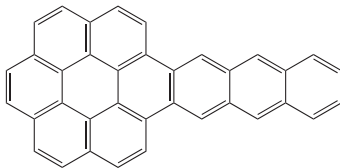
(a) Benzo[ghi]perylene
 $C_{22}H_{12}$



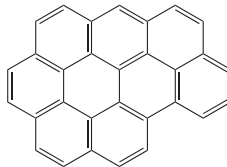
(b) Benzo[a]coronene
 $C_{28}H_{14}$



(c) Naphtho[2,3a]coronene
 $C_{32}H_{16}$



(d) Anthra[2,3a] coronene
 $C_{36}H_{18}$



(d) Naphtho[8,1,2-abc] coronene
 $C_{30}H_{14}$

Polycyclic aromatic hydrocarbons (PAHs). Hydrogen atoms are not shown.

P. Bhattacharyya, D. K. Rai, A. Shukla, *J. Phys. Chem. C* **124**, 14297–14305 (2020).

First Vertical Excitation Energies (HF-based)

Table: First vertical excitation energies (in eV) of selected PAHs computed using the PPP Hamiltonian with different screening parameterizations based on HF orbitals. MRSDCI (Multi-Reference Singles and Doubles Configuration Interaction) reference values are taken from Ref. [1].

Molecule	MRSDCI*	EOM-pCCD+S (HF)			EOM-CCSD (HF)			Exp
		Screen	Standard	Topology	Screen	Standard	Topology	
C22H12	3.39	3.17	3.47	2.84	2.64	2.97	2.49	–
C28H14	3.42	2.99	3.36	2.75	2.46	2.80	2.30	3.38
C30H14	2.95	2.80	3.19	2.46	2.32	2.65	2.17	3.02
C32H16	2.93	2.88	3.22	2.46	2.42	2.77	2.62	–
C36H18	2.70	2.53	3.05	2.13	2.39	2.73	1.99	–
C24H12	–	3.07	3.45	2.87	2.46	2.77	2.31	2.87
C10H8	–	4.35	4.37	4.29	3.40	3.70	3.25	3.95
C6H6	–	5.33	5.34	5.63	4.03	4.32	3.81	4.72

Statistical Error Analysis (HF-based)

Table: Statistical error analysis (in eV) of excitation energies with respect to MRSDCI reference values. ME = Mean Error, MAE = Mean Absolute Error, SD = Standard Deviation. Only systems with available MRSDCI data were included.

Method	Scheme	ME	MAE	SD
EOM-pCCD+S	Screen	-0.204	0.204	0.126
	Standard	+0.180	0.204	0.150
	Topology	-0.550	0.550	0.070
EOM-CCSD	Screen	-0.632	0.632	0.219
	Standard	-0.294	0.306	0.221
	Topology	-0.764	0.764	0.266

- EOM-pCCD+S is more accurate and consistent than EOM-CCSD; Screen and Standard are closest to reference.
- Topology performs poorly, strongly underestimating energies.

First Vertical Excitation Energies (pCCD-based)

Table: First vertical excitation energies (in eV) of selected PAHs computed using the PPP Hamiltonian with different screening schemes based on pCCD orbitals. MRSDCI reference values are taken from Ref. [1].

Molecule	MRSDCI*	EOM-pCCD+S (pCCD)			EOM-CCSD (pCCD)			Exp
		Screen	Standard	src+loop	Screen	Standard	src+loop	
C22H12	3.39	3.44	3.63	3.50	2.64	2.98	2.48	–
C28H14	3.42	3.30	3.50	3.52	2.47	2.80	2.31	3.38
C30H14	2.95	3.13	3.34	3.31	2.33	2.65	2.20	3.02
C32H16	2.93	3.23	3.43	3.37	2.44	2.77	2.28	–
C36H18	2.70	2.97	3.27	2.94	2.40	2.74	2.07	–
C24H12	–	3.34	3.50	3.59	2.47	2.79	2.32	–
C10H8	–	4.36	4.38	4.33	3.41	3.70	3.25	–
C6H6	–	5.33	5.34	4.88	4.03	4.32	3.81	–

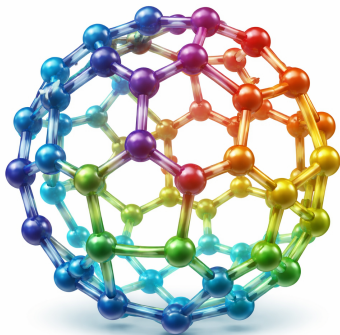
Statistical Error Analysis (pCCD-based)

Table: Statistical error analysis (in eV) of excitation energies with respect to MRSDCI reference values. ME = Mean Error, MAE = Mean Absolute Error, SD = Standard Deviation. Only systems with available MRSDCI data were included.

Method	Scheme	ME	MAE	SD
EOM-pCCD+S	Screen	+0.136	0.184	0.155
	Standard	+0.356	0.356	0.177
	src+loop	+0.250	0.250	0.134
EOM-CCSD	Screen	-0.622	0.622	0.221
	Standard	-0.290	0.306	0.223
	src+loop	-0.810	0.810	0.180

- EOM-pCCD+S with Screen gives the best agreement with MRSDCI (lowest MAE), followed by src+loop, while Standard overestimates more strongly.
- For EOM-CCSD, Standard is closest to MRSDCI, while Screen and src+loop show larger systematic underestimation.

C₆₀ as a Benchmark System



- Highly symmetric π -conjugated system
- Contains different local environments (pentagons and hexagons)
- Strong test for topology-dependent screening

C₆₀ Benchmark: Excitation Energies

- Compared PPP-based methods with SAC-CI (Symmetry-Adapted Cluster Configuration Interaction) reference
- Tested three parameter sets:
 - Standard ($\kappa = 1$)
 - Screened ($\kappa = 2$)
 - Topology-dependent screening
- Method: EOM-CCSD

Key question: Which parametrization best reproduces SAC-CI results?

C₆₀ Benchmark: Key Results

State	SAC-CI	Topology	Screened	Standard
1^1G_g	1.75	1.99	2.15	2.50
1^1T_{2g}	1.90	2.21	2.32	2.50
1^1H_g	2.09	2.25	2.45	–
1^1T_{2u}	2.98	2.44	2.60	–

- Topology screening gives best agreement with SAC-CI
- Standard parameters significantly overestimate energies

R. Fukuda and M. Ehara, *J. Chem. Phys.* **137**, 134304 (2012).

Conclusions & Outlook

- **Benchmark agreement:** pCCD orbitals with screened parameters give the best agreement with MRSDCI (Multi-Reference Singles and Doubles CI) and available experimental data.
- **Screening effects:** For PAHs, screened/standard schemes perform reliably, while topology-based screening leads to systematic underestimation.
- **C₆₀ insight:** In contrast, topology screening provides the best agreement with SAC-CI and correctly reproduces state ordering, highlighting system-dependent screening effects.

References



P. Bhattacharyya, D. K. Rai, and A. Shukla, *J. Phys. Chem. C*, **124** (2020), 14297–14305.



R. Fukuda and M. Ehara, *Electronic excitations of C_{60} fullerene calculated using the ab initio cluster expansion method*, *J. Chem. Phys.*, **137** (2012), 134304.



M. Chandross and S. Mazumdar, *Phys. Rev. B*, **55** (1997), 1497.

*Thank
you!*